

Model of conventional (symmetric key) encryption

- Consider an encryption scheme consisting of
 - ▶ the set of encryption transformations $\{E_e : e \in K\}$
 - ▶ the set of corresponding decryption transformations $\{D_d : d \in K\}$, where K is the key space.
- The encryption scheme is said to be S -key or symmetric-key, if for each associated encryption/decryption key pair (e, d) , it is computationally “easy” to determine d from e and to determine e from d .
- In most practical symmetric-key encryption schemes, $e = d$.
- Other terms used are single-key, one-key, private-key and conventional encryption.

Symmetric-Key Encryption

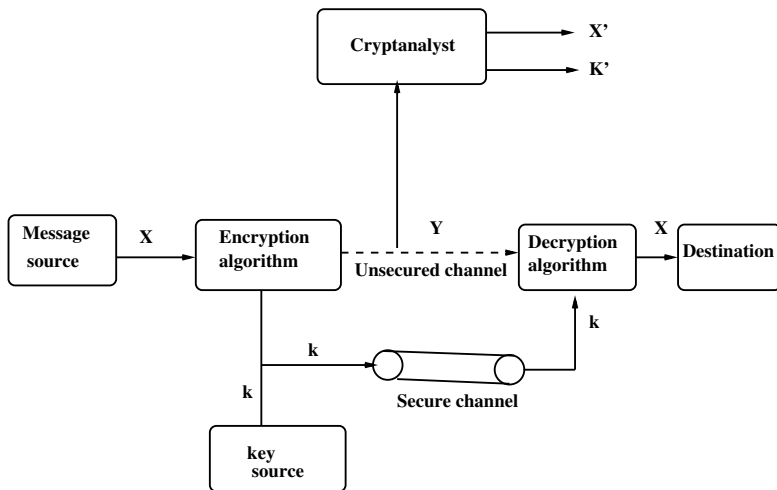


Figure: Model of conventional encryption

Caesar Cipher

- It is the earliest known use of a substitution cipher, and the simplest, was by Julius Caesar.
- Each letter of the alphabet is replaced with the letter standing the three places further down the alphabet.
- For example,
plaintext: meet me after the new year party
ciphertext: PHHW PH DIWHU WKH QHZ BHDU SDUWB
- Each letter is wrapped around, so that the letter following Z is A. Define the transformation by listing all possibilities as follows.

plaintext:	a	b	c	...	v	w	x	y	z
ciphertext:	D	E	F	...	Y	Z	A	B	C

Caesar Cipher

- Encoding technique:

Let us assign a numerical equivalent to each letter:

a	b	c	...	v	w	x	y	z
0	1	2	...	21	22	23	24	25

- Mathematical model:
 - ▶ Encryption: For each plaintext letter p , substitute the ciphertext letter c : $c = E_k(p) = (p + 3) \pmod{26}$, where $k = 3$.
 - ▶ Decryption: For each ciphertext letter c , substitute the plaintext letter p : $p = D_k(c) = (c - 3) \pmod{26}$, where $k = 3$.

The Generalized Caesar Cipher

- A shift may be of any amount, so that the general Caesar algorithm is as follows.
- Mathematical model
 - ▶ Encryption: For each plaintext letter p , substitute the ciphertext letter c : $c = E_k(p) = (p + k) \pmod{26}$, where $0 \leq k \leq 25$.
 - ▶ Decryption: For each ciphertext letter c , substitute the plaintext letter p : $p = D_k(c) = (c - k) \pmod{26}$, where $0 \leq k \leq 25$.

Security issues of the Caesar cipher

- If it is known that a given ciphertext is a Caesar cipher, then a brute-force cryptanalysis is easily performed.
- The key space K in this case contains 25 keys, that is $|K| = 25$.
- Attacker simply tries all the 25 possible keys.
- In this case, the attacker could be able to recover the plaintext as well as the encryption key k from the ciphertext easily (It is an example of Ciphertext-only attack (COA)).

Vernam Cipher

- An encryption system was introduced by an AT& T engineer named Gilbert Vernam in 1918.
- He introduced a new parameter (keyword) which is as long as the plaintext and has no statistical relationship to it.

- **Encryption algorithm**

The system can be expressed as follows:

$$c_i = p_i \oplus k_i$$

where $p_i = i^{th}$ binary digit of plaintext,

$c_i = i^{th}$ binary digit of ciphertext,

$k_i = i^{th}$ binary digit of key,

$\oplus$ = bitwise exclusive-or (XOR) operator.

- **Decryption algorithm**

Because of the properties of XOR, decryption simply involves the same bitwise operation: $p_i = c_i \oplus k_i$.

Vernam Cipher

- **Construction of key:**

- ▶ Keyword should be as long as the plaintext and can be repeating.

- Vernam cipher is an example of classical stream cipher.
- It is also called one-time pad, because each plaintext is appended with random key.
- It is proved in the literature that one-time pad is unbreakable (proof will be given mathematically later), since it produces random output that bears NO statistical relationship to the plaintext.

Vernam Cipher

Problems with the one-time pad

- Generation of key.
- Problem of key distribution and protection.

Because of these difficulties, the one-time is of limited utility, and is used primarily for low-bandwidth channels requiring very high security.